# Chapter

# Modern Physics

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**1. A nucleus of an atom consists of 146 neutrons and 95 protons. It decays after emitting an alpha particle. How many protons and neutrons are left in the nucleus after an alpha emission?**

(a) protons = 93, neutrons = 142

(b) protons = 95, neutrons = 144

(c) protons = 93, neutrons = 144

(d) protons = 95, neutrons = 142

**SQP2025**

**2. Assertion :** Infrared radiations travel long distance through a dense fog and mist.

**Reason :** Infrared radiations undergoes minimal scattering in earth's atmosphere.

(a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(b) Both Assertion and Reason are True and Reason is not the correct explanation of Assertion.

(c) Assertion is false but reason is true.

(d) Assertion is true but reason is false.

**3. (a) Name the radiations that are emitted during the decay of a nucleus, which has highest penetrating power?**

**(b) Does the emission of the above-mentioned radiation result in a change in the mass number?**

**SQP2025**

4. **A radioactive nucleus X emits an alpha particle followed by two beta particles to form nucleus Y.**

**(a) With respect to the element X, where would you position the element Y in the periodic table?**

**(b) What is the general name of the elements X and Y?**

**(c) If the atomic number of Y is 80 then what is the atomic number of X?**

**SQP2025**

5. **What mass of ice at 0°C added to 2.1 kg water, will cool it down from 75°C to 25°C? Given Specific heat capacity of water = 4.2 $Jg^{-1}$ $°C^{-1}$, Specific latent heat of ice = 336 $Jg^{-1}$.**

**SQP2025**

6. **The graph (fig A) illustrates the correlation between the number of protons ($x$-axis) and the number of neutrons ($y$-axis) for elements A, B, C, D and E in the periodic table. These elements are denoted by the letters rather than their conventional symbols. When the element C, depicted in the graph, undergoes radioactive decay, it releases radioactive rays. When these rays are directed into the plane of the paper in the presence of a magnetic field, as indicated in the fig B, they experience deflection, causing them to move upwards.**

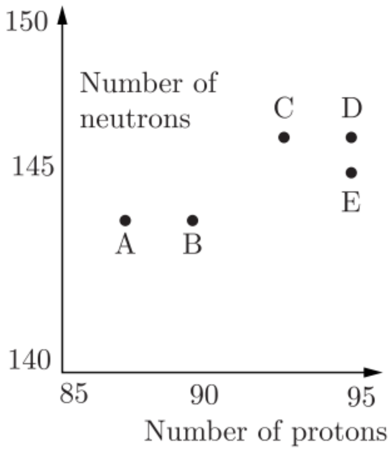


Figure (A)

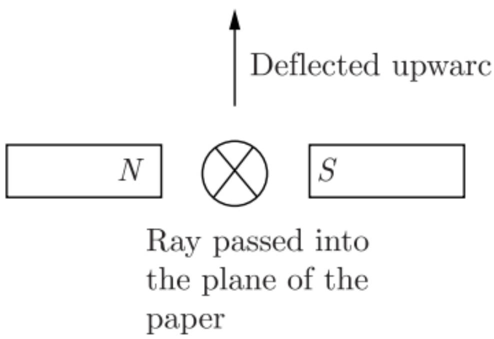


Figure (B)

**(i) Name the radioactive radiations emitted by the element C.**

**(ii) Identify the daughter element from the graph.**

**(iii) Name the law used to identify the radioactive radiations emitted by the element.**

**SQP2025**

**7. A radioactive nucleus containing 128 nucleons emits a 13-particle. After 13-emission the number of nucleons present in the nucleus will be**

(a) 128 (b) 129

(c) 124 (d) 127

**MAIN 2024**

**8. A radioactive element is placed in an evacuated chamber. Then the rate of radioactive decay will**

(a) Decrease

(b) Increase

(c) Remain unchanged

(d) Depend on the surrounding temperature

**MAIN 2024**

**9. Complete the following radioactive reaction:**

$$\mathrm{X} \rightarrow \mathrm{Y} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{234}_{91}\mathrm{Z} + {}^{0}_{-1}\mathrm{e}$$

**MAIN 2024**

**10. Uranium is available in two forms U-235 and U-238. Which of the two isotopes of Uranium is more fissionable ?**

**MAIN 2024**

**11. State one factor that affects the magnitude of induced current in an AC generator.**

**MAIN 2024**

**12. In the following atoms, which one is a radioisotope? Give one use of this isotope. $O^{16}$, $C^{14}$, $N^{14}$, $He^{4}$**

**MAIN 2024**

**13. Copy and complete the nuclear reaction by filling in the blanks.**

$$_{92}\mathrm{U}^{235} + {}_{0}\mathrm{n}^{1} \longrightarrow {}_{56}\mathrm{Ba}^{\cdots} + {}_{\ldots}\mathrm{Kr}^{92} + 3\,{}_{0}\mathrm{n}^{1}$$

**MAIN 2024**

**14. Below is an incomplete table showing the arrangement of electromagnetic spectrum in the increasing order of their wavelength. Complete the table.**

| Gamma ray | X-ray | U V rays | Visible rays | Infrared | A | Radio |
|---|---|---|---|---|---|---|

**(a) Identify the radiation A.**

**(b) Name the radiation used to detect fracture in hones.**

**(c) Name one property common to both A and Radio waves.**

**MAIN 2024**

**15. The Sun radiates energy in all directions. The average radiation received on the earth surface from the Sun 1.4 kW/m$^2$. The average Earth-Sun distance 1.5 X 10$^{11}$ m. The mass lost by the Sun per day is (1 day = 86400 s)**

(a) 4.4 X 10$^9$ kg

(b) 3.8 X 10$^{14}$ kg

(c) 3.8 X 10$^{12}$ kg

(d) 7.6 X 10$^{14}$ kg

**COMP 2023**

**16. Why are they generally not able to leave the metallic surface? SQP 2022**

**17. The heaviest nuclear radiation is**

(a) $X$-radiation

(b) $\alpha$-radiation

(c) $\gamma$-radiation

(d) $\beta$-radiation

**SEM-II 2022**

**18. To study the age of excavated material of archaeological significance we study the rate of decay of an isotope of**

(a) Uranium

(b) Cobalt

(c) Carbon

(d) Chlorine

**SEM-II 2022**

**19. (i) What is nuclear energy?**

**(ii) After emission of a nuclear radiation, the atomic number of the daughter nucleus increases by 1. Identify the nuclear radiation.**

**(iii) Write a nuclear reaction indicating the nuclear change mentioned in (ii).**

**(iv) What is the special name given to the parent and daughter nucleus when this radiation is emitted?**

**SEM-II 2022**

**20. Complete and rewrite the following nuclear reaction by filling the blanks.**

$$^{235}_{\ldots}\mathrm{U} \longrightarrow {}^{\ldots}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

**SEM-II 2022**

**21. Suggest one way by which these electrons could be made to leave the metal surface.**

**COMP 2022**

**22. In fission of one uranium-235 nucleus, the loss in mass is 0.2 amu. Calculate the energy released.**

(a) 156.4 MeV

(b) 182.6 MeV

(c) 186.2 MeV

(d) 189.1 MeV

**MAIN 2022**

**23. In nuclear reaction**

$$^{27}_{12}\mathrm{Mg} \xrightarrow{-\beta} \mathrm{Al} \xrightarrow{-\gamma}$$

(a) $^{27}_{12}$Mg emits a $\beta$-particle and is transformed to aluminium. What is the mass number and the atomic number of aluminium?

(b) Aluminium emits a $\gamma$ ray. What is the resulting nucleus?

**MAIN 2022**

**24. When an element gives out high energy radiations on its own, the change which takes place is:**

(a) nuclear change (b) chemical change

(c) physical change (d) none of these

**MAIN 2021**

**25. A certain radioactive nucleus emits a particle that leaves its mass unchanged but increases its atomic number by one. Identify the particle and write its symbol.**

**MAIN 2021**

**26. What do you mean by a radiograph ?** **COMP 2021**

**27. What is background radiation? How can you detect background radiation?**

**COMP 2021**

# Solution

**1.**

**Ans: (c)** protons = 93, neutrons = 144

**Explanation:**

An alpha particle is a helium nucleus: **2 protons + 2 neutrons**.

Initial nucleus:

- Protons = 95
- Neutrons = 146

After emitting 1 alpha particle:

- Protons = 95 − 2 = **93**
- Neutrons = 146 − 2 = **144**

So, the nucleus left has **93 protons and 144 neutrons**.

**2.**

**Ans:** Infrared radiations can travel long distances through dense fog and mist because they undergo very little scattering in the Earth's atmosphere.

Therefore, both the Assertion and the Reason are true, but the Reason is **not** the correct explanation of the Assertion.

Hence, the correct option is **(b)**.

**3.**

**Ans:**

**(a)** *Gamma (γ) radiations.*

**(b)** *No, the emission of gamma radiation does not cause any change in the mass number.*

**4.**

**Ans:**

**(a)** *At the same place in the periodic table.*

**(b)** *They are called isotopes.*

**(c)** *The atomic number of X is 80.*

**5.**

**Ans:**

Using the principle of calorimetry:

**Heat lost = Heat gained**

$$mc\Delta t = m_1 L + m_1 c\Delta t$$

For water cooling from 75 $°C$ to 25 $°C$:

$$2100 \times 4.2 \times (75 - 25)$$

For ice melting and warming from 0° $C$ to 25° $C$:

$$m \times 336 + m \times 4.2 \times (25 - 0)$$

Equating:

$$2100 \times 4.2 \times 50 = m \times (336 + 105)$$

$$2100 \times 50 = 105\,m$$

$$m = \frac{2100 \times 50}{105}$$

$$m = 1000\text{g}$$

**Mass of ice required = 1000 g (i.e., 1 kg).**

**6.**

**Ans:**

**(i)** Beta

**(ii)** Element E

**(iii)** Fleming's left-hand rule

**7.**

**Ans: (a)** 128

**Explanation:**

In **β-emission**, a neutron in the nucleus converts into a proton and an electron (β-particle).

- The **atomic number increases by 1** (because one more proton).
- The **mass number (total nucleons = protons + neutrons) remains the same**, since a neutron is simply changed into a proton.

Initial nucleons = 128 → After β-emission, still **128 nucleons**.

**8.**

**Ans: (c)** Remain unchanged

**Explanation:**

Radioactive decay is a **nuclear** process. It depends only on the nature of the unstable nucleus, not on external conditions like pressure, temperature, or whether the chamber is evacuated. So, placing the element in an evacuated chamber does **not** affect its rate of decay.

**9.**

**Ans:**

$$^{234}_{90}\mathrm{X} \rightarrow {}^{230}_{88}\mathrm{Y} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{234}_{91}\mathrm{Z} + {}^{0}_{-1}\mathrm{e}$$

**10.**

**Ans:** U-235.

**11.**

**Ans:** Number of turns in armature may be increased.

**12.**

**Ans:** $C^{14}$ is a radioisotope because of the instability of the nucleus. It's used for carbon dating.

**13.**

**Ans:**

$${}_{92}\mathrm{U}^{235} + {}_{0}\mathrm{n}^{1} \rightarrow {}_{46}\mathrm{Ba}^{141} + {}_{36}\mathrm{Kr}^{92} + 3{}_{0}\mathrm{n}^{1}$$

**14.**

**Ans:**

**(a)** The radiation **A** is **Microwaves**.

**(b)** The radiation used to detect fractures in bones is **X-rays**.

**(c)** One property common to both **A (microwaves)** and **radio waves** is:
**Both travel with the same speed in vacuum (speed of light).**

**15.**

**Ans: (b)** $3.8 \times 10^{14}$ kg

**Explanation:**

The Sun radiates energy in all directions, and the average energy received on Earth is **1.4 kW/m²**.

Using Einstein's equation $E = mc^2$, the total energy radiated by the Sun per day is converted into mass loss.

Hence, the **mass lost by the Sun per day** is calculated as

$$m = \frac{E}{c^2} = \frac{4\pi(1.5 \times 10^{11})^2 \times 1.4 \times 10^3 \times 86400}{(3 \times 10^8)^2} \approx 3.8 \times 10^{14}\ \mathrm{kg}$$

Thus, the Sun loses about **$3.8 \times 10^{14}$ kg** of mass every day.

**16.**

**Ans:** At ordinary temperature, the free electrons in a metal cannot leave its surface because they are held by attractive surface forces. The energy they possess is not sufficient to overcome these forces, so they cannot escape from the metal surface.

**17.**

**Ans: (b)** α-radiation

**Explanation:**

Among all types of nuclear radiations:

- **α-particles** are the **heaviest**, consisting of **2 protons and 2 neutrons** (a helium nucleus).
- **β-particles** are light electrons, and
- **γ-rays** and **X-rays** are massless electromagnetic waves.

Hence, **α-radiation** is the heaviest type of nuclear radiation.

**18.**

**Ans: (c)** Carbon

**Explanation:**

To find the age of archaeological samples (bones, wood, charcoal, etc.), scientists use **Carbon-14 dating**. Carbon-14 is a radioactive isotope of carbon that decays at a known, constant rate. By measuring how much Carbon-14 remains in a sample, the time since the organism died can be estimated.

So, we study the decay of an isotope of **carbon**, hence option **(c)**.

**19.**

**Ans:**

**(i)** Nuclear energy is the energy released during the transformation of one nucleus into another, i.e., during nuclear reactions.

**(ii)** The radiation that increases the atomic number of the daughter nucleus by 1 is **beta (β) radiation**.

**(iii)** The nuclear reaction for β-decay is:

$$ {}_0^1n \rightarrow {}_1^1p + {}_{-1}^{0}e $$

**(iv)** In this process, the **parent nucleus** is called a **neutron**, and the **daughter nucleus** formed is a **proton**.

**20.**

**Ans:** The general equation for α-decay is:

$$ {}_Z^AX \longrightarrow {}_{Z-2}^{A-4}Y + {}_2^4\mathrm{He} $$

For the given reaction:

$$ {}_x^{235}\mathrm{U} \longrightarrow {}_{90}^{231}\mathrm{Th} + {}_2^4\mathrm{He} $$

Comparing atomic numbers:

$$ x = 90 + 2 = 92 $$

Comparing mass numbers:

$$ 235 = y + 4 \quad \Rightarrow \quad y = 235 - 4 = 231 $$

Thus, the **complete nuclear reaction** is:

$$\boxed{{}^{235}_{92}\mathrm{U} \longrightarrow {}^{231}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}}$$

**21.**

**Ans:** Electrons can be made to leave the metal surface by giving them extra energy from an external source such as heat, light, or a strong electric field.

**22.**

**Ans: (c)** 186.2 MeV

**Explanation:**

Energy released in nuclear reactions is given by

$$E = \Delta mc^2$$

and 1 $amu$ = 931.5 MeV.

Given loss of mass $\Delta m$ = 0.2 $amu$:

$$E = 0.2 \times 931.5 \approx 186.3 \text{ MeV} \approx 186.2 \text{ MeV}$$

So, the energy released is **186.2 MeV**.

**23.**

**Ans:**

(a) ${}^{27}_{12}\mathrm{Mg} \xrightarrow{-\beta} {}^{27}_{13}\mathrm{Al}$

Mass number of aluminium = 27
Atomic number of aluminium = 13

(b) ${}^{27}_{13}\mathrm{Al} \xrightarrow{-\gamma}$ No change in the nucleus

**24.**

**Ans: (a)** Nuclear change

**Explanation:**

When an element emits high-energy radiations on its own, the change happens inside the **nucleus** of the atom. Such a process is called **radioactive decay**, which is a **nuclear change**. In this process, one element may transform into another due to changes in the nucleus, releasing a large amount of energy.

**25.**

**Ans:** The particle is a **beta (β) particle**, and it is represented by the symbol:

$$ {}^{0}_{-1}e $$

**26.**

**Ans:** Radiograph is the photograph of the bones inside the body on a photographic film.

**27.**

**Ans:** Background radiation is the radiation present everywhere on Earth due to radioactive substances in the ground and cosmic rays from space.

It can be detected using a **Geiger–Müller (Geiger) counter**.